Amazon ECS Cluster Setup with EC2 Infrastructure

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon ECS cluster** was created from the **AWS Management Console** and the role of EC2 infrastructure was explained. The video showed that ECS clusters created from the console always include **Fargate** and **Fargate Spot** by default, and it then demonstrated how to add **EC2 infrastructure** during cluster creation. When EC2 is added, AWS automatically creates the required **Auto Scaling Group**, **launch template**, and **ECS-optimized EC2 instance** setup. The lab verified that the EC2 instance becomes part of the ECS cluster infrastructure and can be scaled by adjusting the Auto Scaling Group desired capacity. It also mentioned that if EC2 is not selected during creation, you can still add it later.

Understanding ECS Cluster Creation

An **Amazon ECS cluster** is the logical place where your containers run. When you create a cluster from the AWS Console, ECS gives you built-in serverless options and also lets you add EC2 capacity for more control. The newer UI shows three main infrastructure choices:

- **Fargate only**
- **Fargate and Managed Instances**
- **Fargate and Self-managed instances**

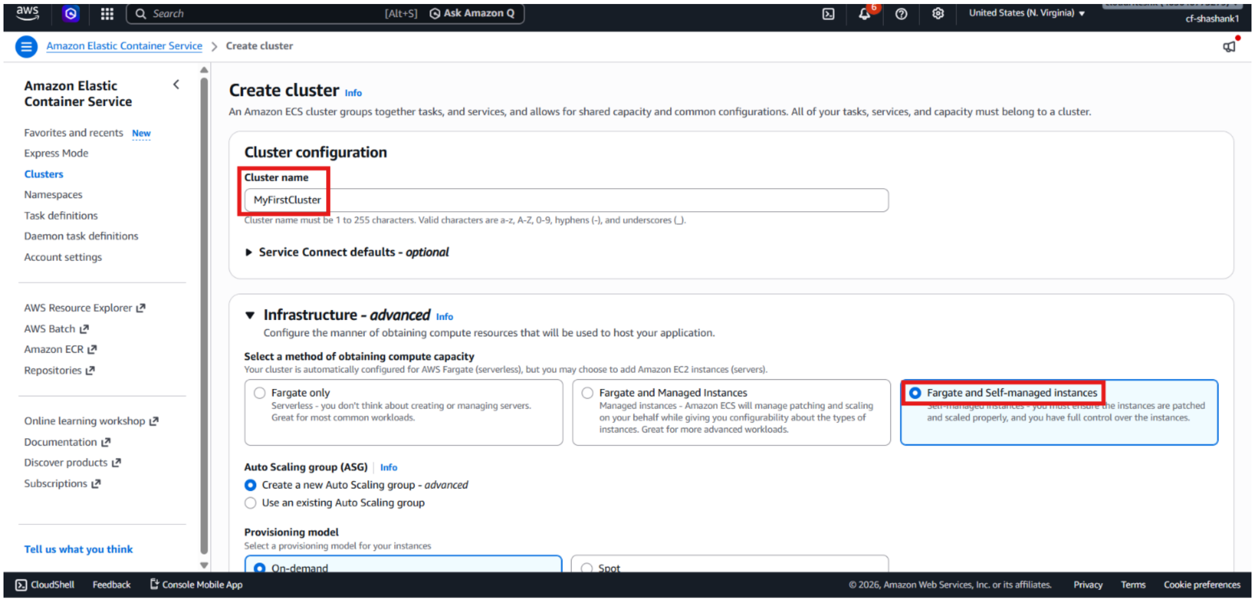
Fargate only is serverless, so AWS manages the compute layer. **Managed Instances** means AWS helps manage the EC2 layer for you. **Self-managed instances** means you control the EC2 instances yourself and attach them to the ECS cluster.

The lab focused on creating a cluster with EC2 added during cluster creation. ECS then created the supporting infrastructure automatically, including the Auto Scaling Group and the ECS EC2 instance registration.

Lab Steps

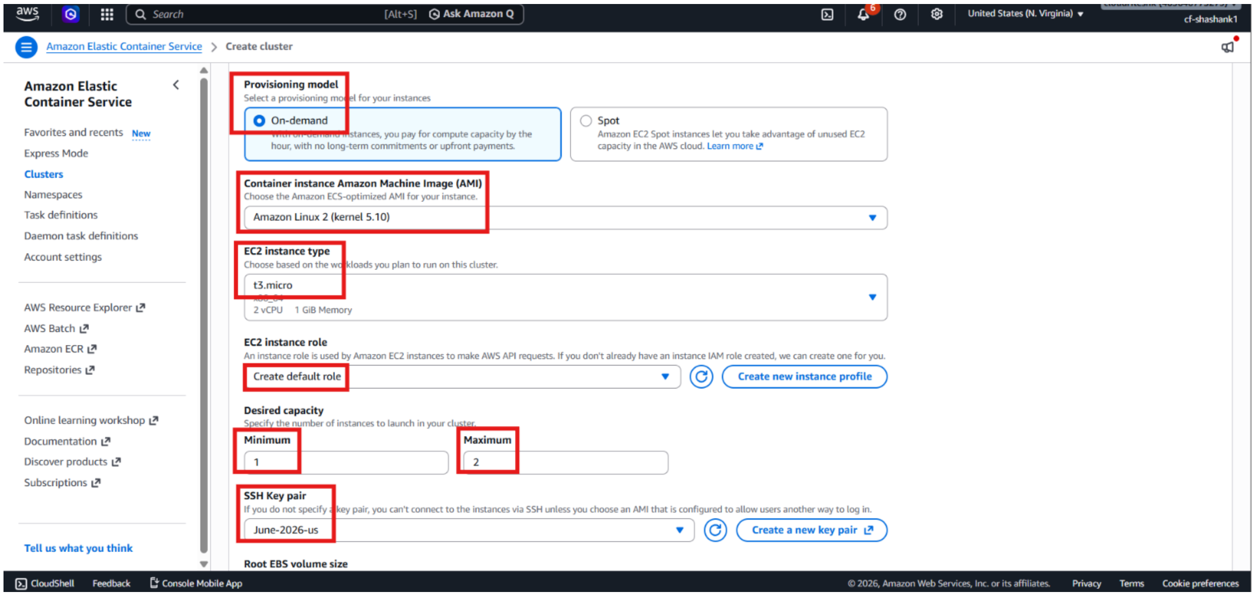
Step 1 – Open ECS and Create a Cluster

- Sign in to the AWS Management Console.
- Open **Amazon ECS**.
- Enter the cluster name: MyFirstCluster
- Review the infrastructure options shown in the current UI:
    - **Fargate only**
    - **Fargate and Managed Instances**
    - **Fargate and Self-managed instances**
- Choose **Fargate and Self-managed instances**

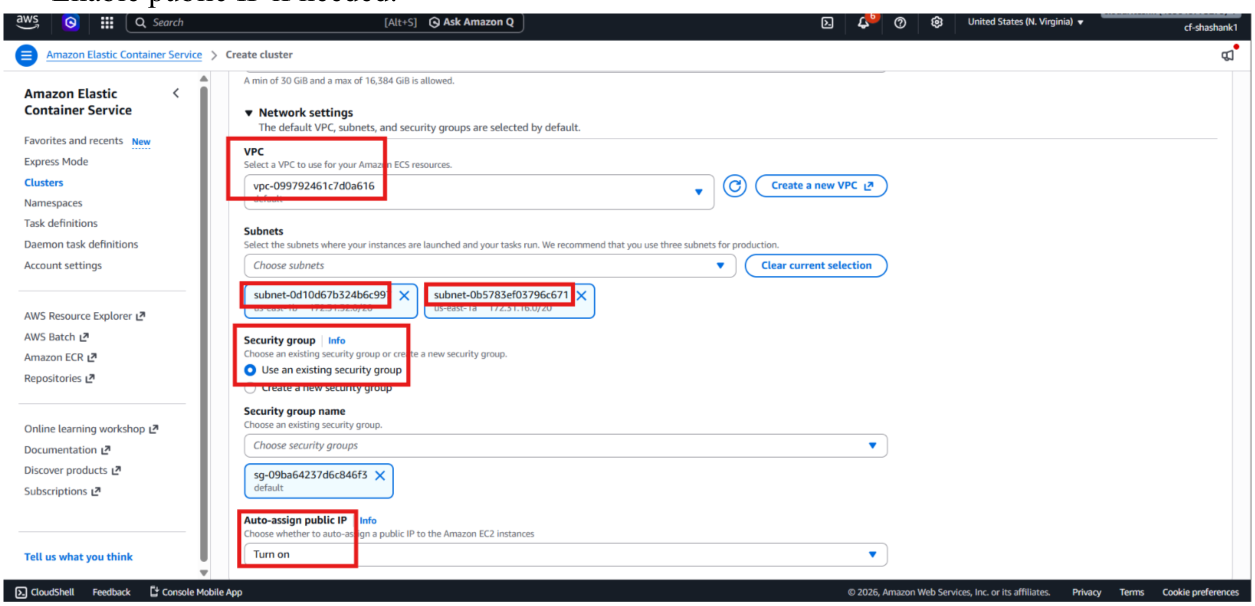


Step 2 – Configure the EC2 Infrastructure

- Choose to create a **new Auto Scaling Group**.
- Select the provisioning model:
  - **On-demand** for predictable usage
  - **Spot** for lower-cost spare capacity
- Choose an **ECS-optimized AMI** and for that i have selected **Amazon Linux 2 (kernel 5.10)**.
- Select an EC2 instance type, such as **t2.micro**.
- Create a new default role for now and attach the EC2 IAM role if you have already created one.
- Provide the desired instance count:
  - Minimum : 1
  - Maximum : 2
- Choose the key pair if already created otherwise create a new one.

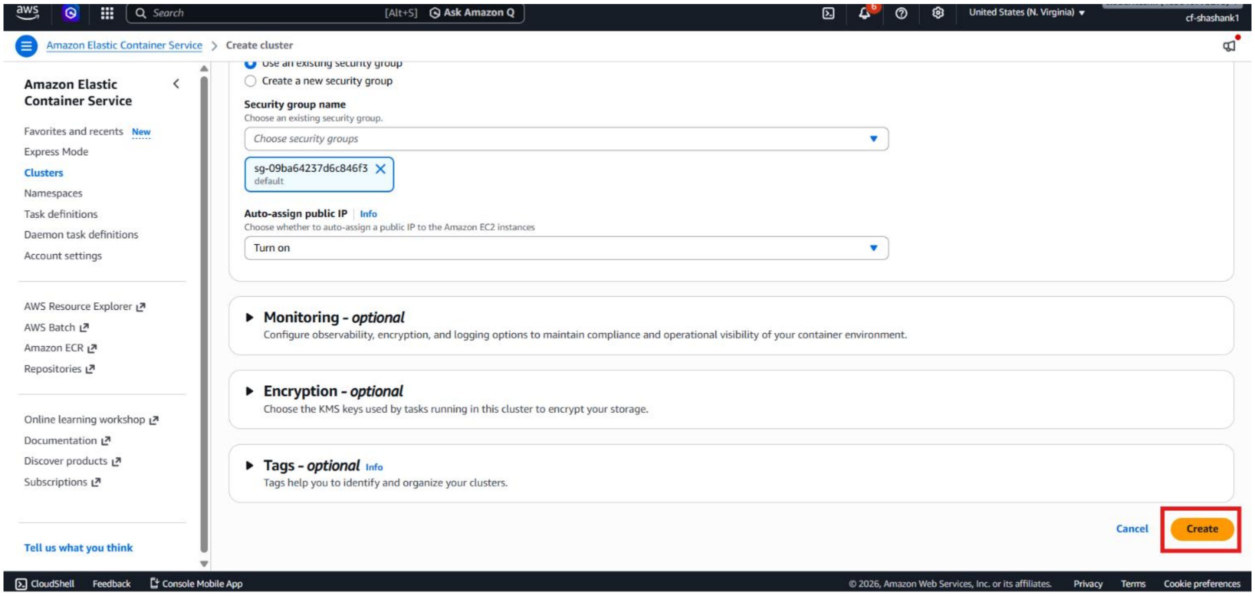


- Select the default VPC and subnets.
- Select the security group.
- Enable public IP if needed.

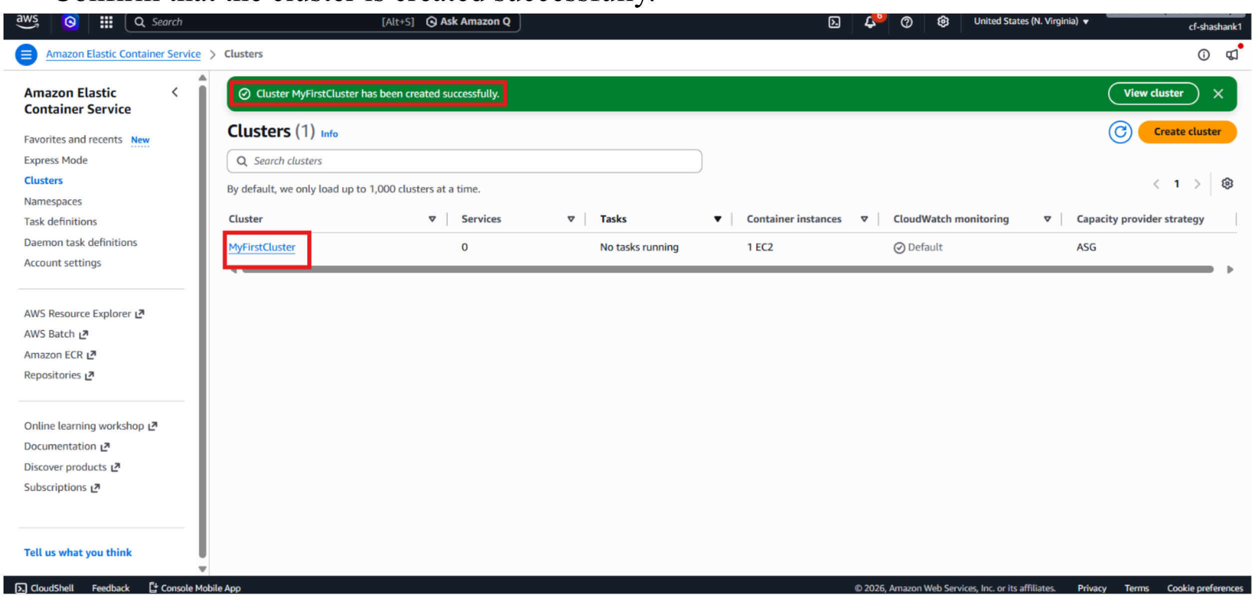


Step 3 – Create the Cluster

- Review the settings.
- Click **Create**.

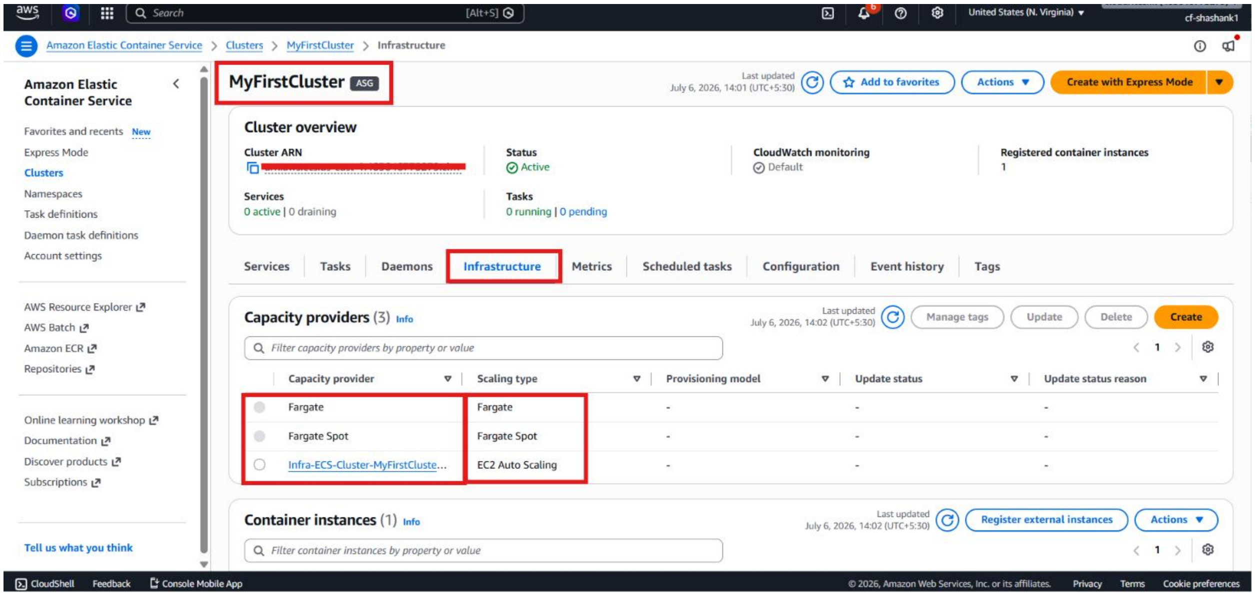


- Wait for ECS to finish provisioning the infrastructure.
- Confirm that the cluster is created successfully.

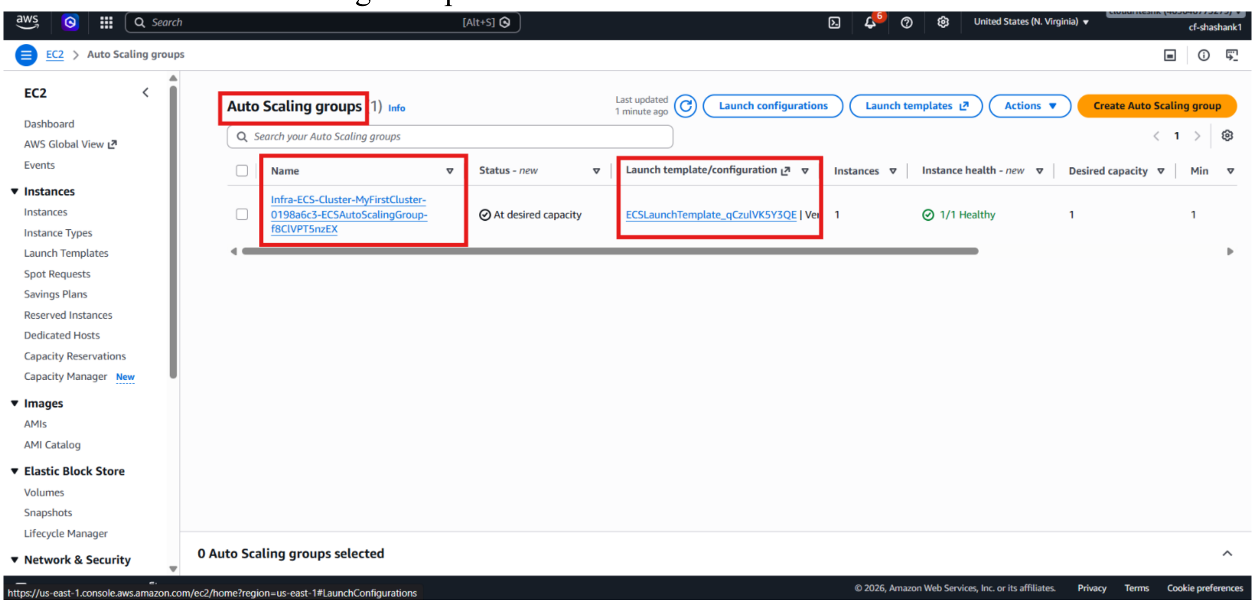


Step 4 – Verify the ECS Infrastructure

- Open the cluster.
- Go to the **Infrastructure** tab.
- Confirm that:
    - Fargate is available
    - Fargate Spot is available
    - The EC2 infrastructure is present



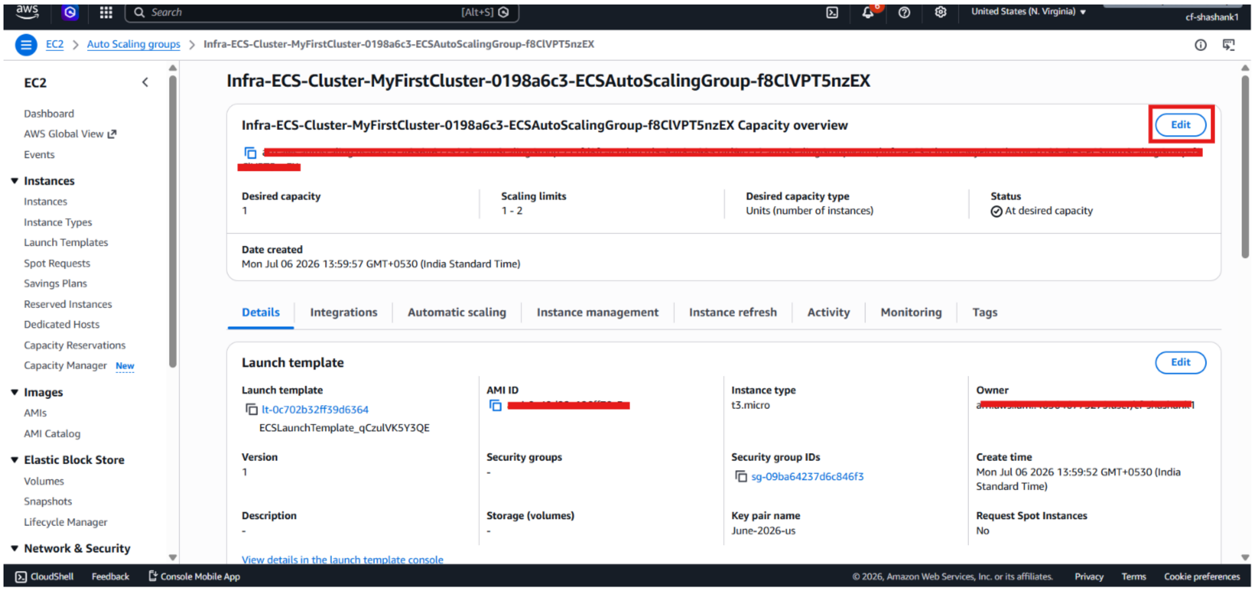
- The Auto Scaling Group exists



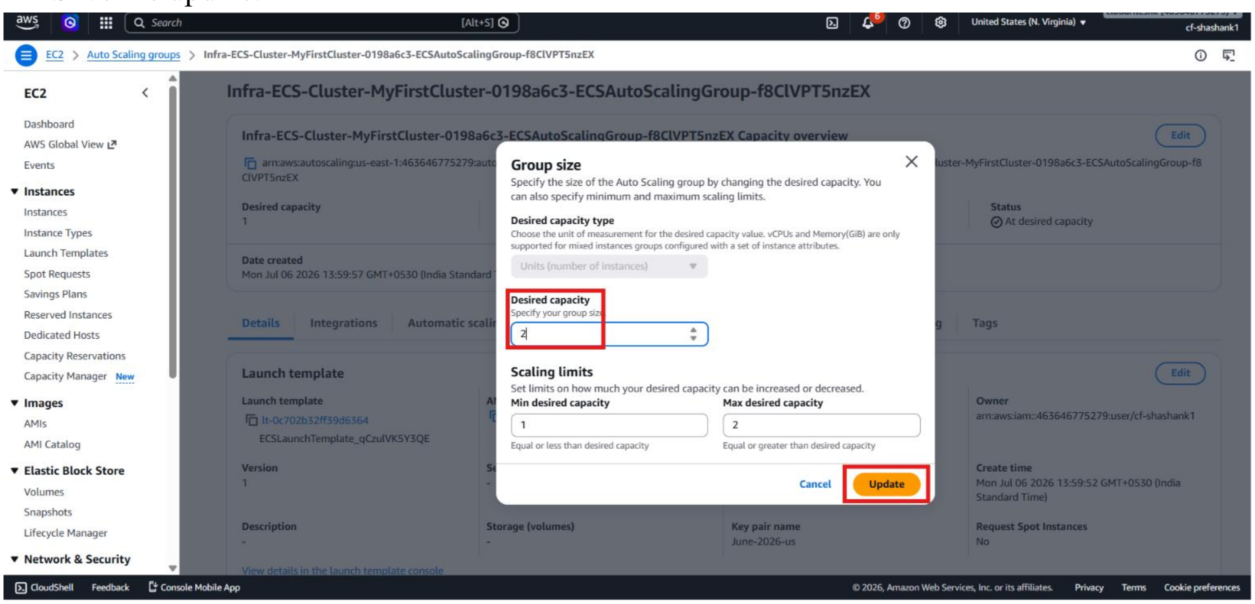
- The EC2 instance is registered in the cluster

Step 5 – Scale the EC2 Capacity

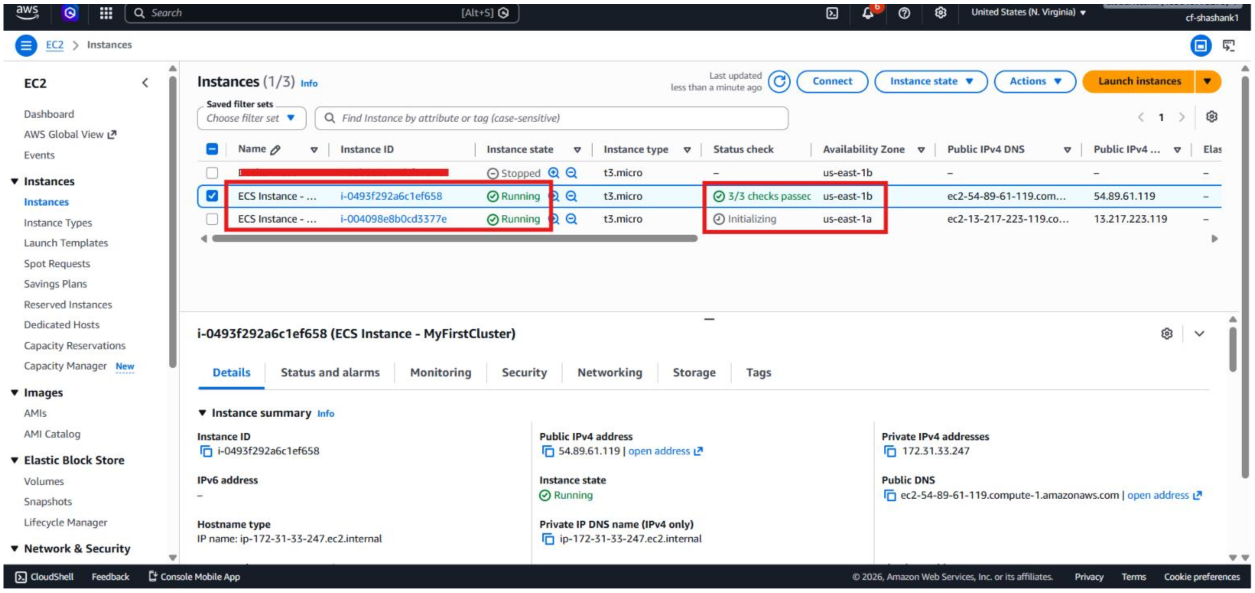
- Open the Auto Scaling Group.
- Edit the desired capacity.



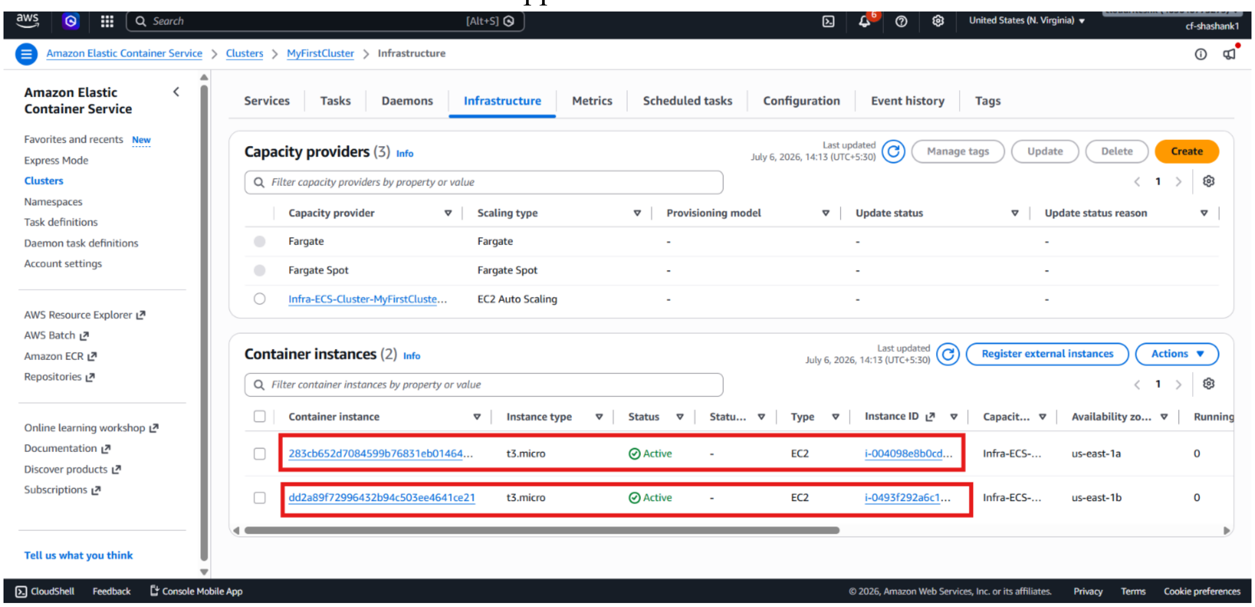
- Increase the number of instances if needed.
- Save the update.



- Wait for the new EC2 instance to launch.



- Refresh the ECS cluster infrastructure tab.
- Confirm that the new EC2 instance appears in the cluster.



Verification

- The ECS cluster was created successfully.
- EC2 infrastructure was added during cluster creation.
- ECS created the Auto Scaling Group automatically.
- The ECS-optimized EC2 instance registered with the cluster.
- The cluster infrastructure tab showed the EC2 instance.
- Increasing the Auto Scaling Group capacity added another instance successfully.